

A faded background image of an elephant standing in front of a wooden fence. The elephant is looking down at a large, orange, spherical object on the ground.

S&DS 265 / 565  
Introductory Machine Learning

# Bayesian Inference in ML

February 25

Yale

# Quick notes

- Quiz 3 live after class
- Midterm week from today
- Material up to and including today
- Will announce mid-semester grades after midterm
- Assn 3 due after mid-semester break

# Midterm review sessions

Ruiqi: Friday, Feb 27, 7:00pm, KT 205

Jianliang: Saturday, Feb 28, 5:00pm, TBA

Dingyao: Monday, March 2, 8:00pm, KT 205

John: Tuesday, March 3, 5:30pm, KT 205

# Reminder

- If you need to email me about any issues with your work for the course, please email [sds265@yale.edu](mailto:sds265@yale.edu)
- Otherwise, your email may be missed

# For Today

- Mixtures
- Introduction to Bayesian analysis
- Overview of topic models

# Where are we going?

- So far: Mostly “surface representations” & explicit features
  - ▶ Listing information for CA housing
  - ▶ Supernovæ classification — measured light

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- So far: Mostly “surface representations” & explicit features
  - ▶ Listing information for CA housing
  - ▶ Supernovæ classification — measured light
- Next: “Hidden” representations and latent variables

# George Washington

1789

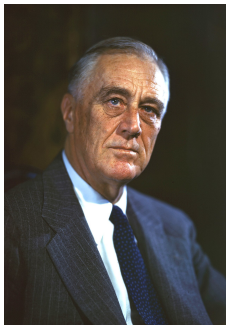


*Among the many interesting objects which will engage your attention that of providing for the common defense will merit particular regard. To be prepared for war is one of the most effectual means of preserving peace.*



# Franklin Roosevelt

1941



*Such a peace would bring no security for us or for our neighbors. As a nation, we may take pride in the fact that we are softhearted; but we cannot afford to be soft-headed.*

# SOTU models



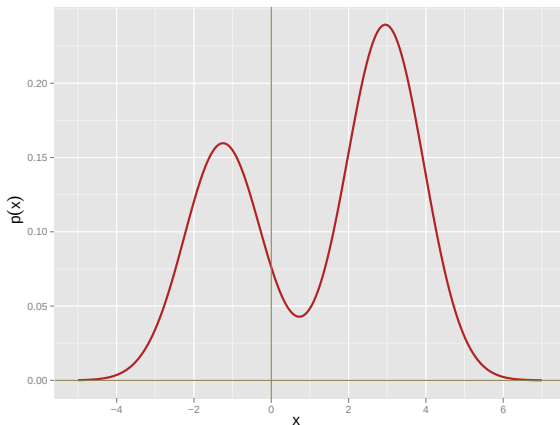
# The elephants in the room



# Mixtures

- Key technique: Mixture models
- Mixtures have latent variables
- Flexible tool
- Simple and difficult at the same time

# Gaussian Mixture



Mixture of two normals, one with mean  $-\frac{5}{2}$ , another with mean 3.

# Mixtures

- For each data point, we have a hidden (latent) variable: Which Gaussian generated the data point?
- $Z = 1$  it was generated from one model;  $Z = 0$  it was generated from the other model
- We don't *observe*  $Z$ , it could be either value

# Mixtures

- *Mixture of  $f$  and  $g$ :*

$$p(x) = \theta f(x) + (1 - \theta)g(x)$$

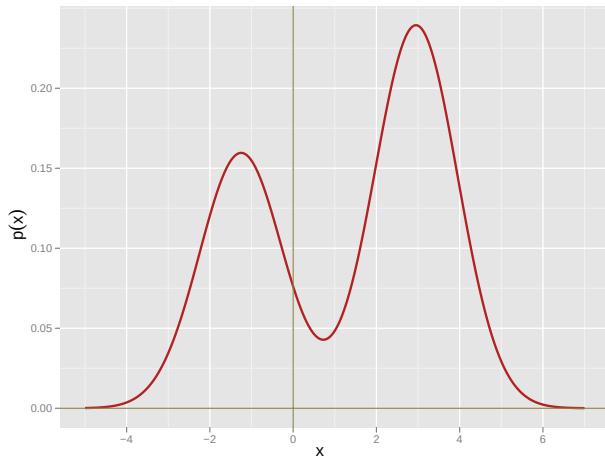
Simplest, most common kind of latent variable model

- *Hidden variable representation:* Define  $Z \sim \text{Bernoulli}(\theta)$  and

$$p(x) = \sum_{z=0,1} p(x | z) p(z)$$

with  $p(x | 1) = f(x)$ ,  $p(x | 0) = g(x)$ ,  $p(z) = \theta^z(1 - \theta)^{(1-z)}$ .

# Gaussian Mixture: All the Key Concepts





# Bayesian Inference

The parameter  $\theta$  of a model is viewed as a random variable.  
Inference usually carried out as follows:

- Choose a *generative model*  $p(x | \theta)$  for the data.
- Choose a *prior distribution*  $\pi(\theta)$  that expresses beliefs about the parameter before seeing any data.
- After observing data  $\mathcal{D}_n = \{x_1, \dots, x_n\}$ , update beliefs and calculate the *posterior distribution*  $p(\theta | \mathcal{D}_n)$ .

# Bayes' Theorem

A simple consequence of conditional probability:

$$\begin{aligned}\mathbb{P}(A|B) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(B|A) \mathbb{P}(A)}{\mathbb{P}(B)}\end{aligned}$$

# Bayes' Theorem

The posterior distribution can be written as

$$p(\theta | x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n | \theta)\pi(\theta)}{p(x_1, \dots, x_n)} = \frac{\mathcal{L}_n(\theta)\pi(\theta)}{c_n} \propto \mathcal{L}_n(\theta)\pi(\theta)$$

where  $\mathcal{L}_n(\theta)$  is the *likelihood function* and

$$c_n = p(x_1, \dots, x_n) = \int \mathcal{L}_n(\theta)\pi(\theta)d\theta$$

is the normalizing constant, which is also called *evidence*.

# Important Example

Take model  $X \sim \text{Bernoulli}(\theta)$ .

This is a “coin flip”:  $X = 1$  means “heads” and  $X = 0$  means “tails.”

Natural prior is  $\text{Beta}(\alpha, \beta)$  distribution

$$\pi_{\alpha, \beta}(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

# Important Example

Take model  $X \sim \text{Bernoulli}(\theta)$ .

Natural prior is  $\text{Beta}(\alpha, \beta)$  distribution

$$\pi_{\alpha, \beta}(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

The scaling constant is scary looking:

$$\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)}$$

where  $\Gamma(\cdot)$  is the “Gamma function”

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx$$

# Important Example

$X \sim \text{Bernoulli}(\theta)$  with data  $\mathcal{D}_n = \{x_1, \dots, x_n\}$ . Prior  $\text{Beta}(\alpha, \beta)$  distribution

$$\pi_{\alpha, \beta}(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Let  $s = \sum_{i=1}^n x_i$  be the number of “heads”

Posterior distribution  $\theta \mid \mathcal{D}_n$  is another beta distribution!

Specifically, with

$$\tilde{\alpha} = \alpha + \text{number of heads} = \alpha + s$$

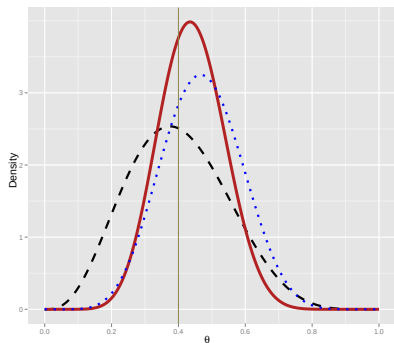
$$\tilde{\beta} = \beta + \text{number of tails} = \beta + n - s$$

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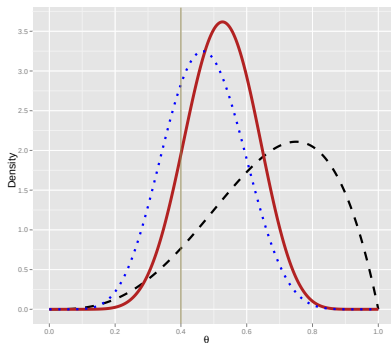
Showing this just uses the simple fact that  $\theta^{\alpha-1} \theta^x = \theta^{x+\alpha-1}$

# Example

$n = 15$  points sampled as  $X \sim \text{Bernoulli}(\theta = 0.4)$ , with  $s = 7$  heads.



prior A



prior B

Prior distribution (black-dashed), likelihood function (blue-dotted),  
posterior distribution (red-solid).

# Dirichlet

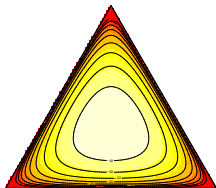
Multinomial model with Dirichlet prior is generalization of the Bernoulli/Beta model.

$$\text{Dirichlet}_{\alpha}(\theta) \propto \theta_1^{\alpha_1-1} \theta_2^{\alpha_2-1} \dots \theta_K^{\alpha_K-1}$$

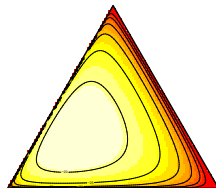
where  $\alpha = (\alpha_1, \dots, \alpha_K) \in \mathbb{R}_+^K$  is a non-negative vector.



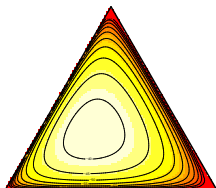
# Example



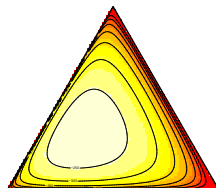
prior with Dirichlet(6,6,6)



likelihood function with  $n = 20$



posterior distribution with  $n = 20$



posterior distribution with  $n = 200$

# Example: Election Forecasting

<https://projects.economist.com/us-2020-forecast/president/how-this-works>



Today

Weekly edition

Menu



## Forecasting the US elections

*The Economist* is analysing polling, economic and demographic data to predict America's elections in 2020

→ [Read more of our election coverage](#)

**President**

Senate

House

National forecast

**How this works**

COMPETITIVE STATES

Arizona

Florida

Georgia

Iowa

Michigan

Nevada

New Hampshire

North Carolina

Ohio

Pennsylvania

Texas

Wisconsin

ALL STATES

Alabama

## How The Economist presidential forecast works

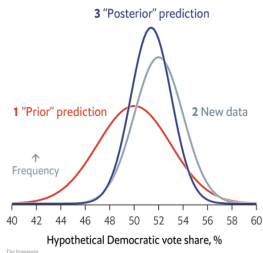
**T**HIS YEAR, *The Economist* is publishing its first-ever statistical forecast of an American presidential election. Developed with the assistance of Andrew Gelman and Merlin Heidemanns, political scientists at Columbia University, our model calculates Joe Biden's and Donald Trump's probabilities of winning each individual state and the election overall. Its projections will be updated every day at <https://projects.economist.com/us-2020-forecast/president>.

In another first, we are [publishing the source code](#) for what we believe to be the most innovative section of the model. All readers are welcome to download it, explore how it works, tweak its parameters and run it

# Example: Election Forecasting

<https://projects.economist.com/us-2020-forecast/president/how-this-works>

## Three steps of Bayesian inference



## Back to Bayes-ics

Readers acquainted with the workings of similar forecasting models may be surprised that the phrase “state polls” has not yet entered the equation. This exclusion is by design. Our model follows a logical structure first developed by Thomas Bayes, an 18th-century reverend whose ideas have shaped a large and growing family of statistical techniques. His approach works in two stages. First, before conducting a study, researchers

explicitly state what they believe to be true, and how confident they are in that belief. This is called a “prior”. Next, after acquiring data, they update this prior to reflect the new information—gaining more confidence if it confirms the prior, and generally becoming more uncertain if it refutes the prior (though not if the new numbers are so definitive that leave little room for doubt). In this framework, the expected distribution of potential vote shares in each state derived above is the prior, and state polls that trickle in during the course of the campaign are the new data. The result—a “posterior”, in Bayesian lingo—is our forecast.

# Let's go to the notebook!

## Demo code for Bayesian analysis

In this notebook we illustrate some of the basic models and priors for Bayesian inference. These concepts will be important for our discussions about "topic models."

First, we illustrate the situation where the parameter  $\theta$  that we are modeling is a Bernoulli parameter. This can be thought of as the probability that flipping a certain coin comes up heads. The most commonly used prior distribution for this model is a beta distribution. Under a beta prior, the posterior distribution is again a beta distribution.

This is illustrated in the following simulation. The prior distribution is a beta distribution. As the sample size increases, the posterior distribution becomes concentrated on the true parameter. But as the variance of the prior distribution increases, the posterior distribution becomes more spread out.

```
In [5]: import os, gzip
import numpy as np
import matplotlib.pyplot as plt
```

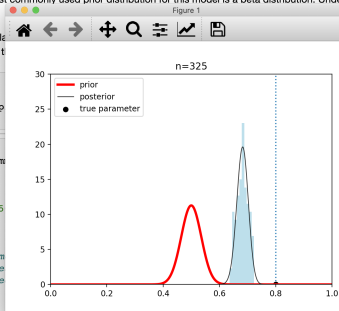
```
In [*]: %matplotlib qt
from scipy.special import gamma
from scipy import random
from scipy.stats import beta

theta = np.linspace(0,1,num=5)
fig = plt.figure(1)
plt.ion()
```

```
# The following are the parameters
# variance of the prior decreases
# the posterior to be centered
```

```
scale = 100
a0 = scale*1
b0 = scale*1

sample_size = 100
```



# Reading

- See notes on Bayesian inference on iML site  
(`bayes-notes.pdf`)
- Not necessary to understand everything—but you should be able to do some of the basic “coin flipping” calculations
- We'll build on this when discussing topic models

# Summary

- Mixtures are latent variable models
- The mixing weight encodes a hidden variable
- Computing with mixtures uses basic probabilistic reasoning
- Bayesian inference is popular in ML
- In a Bayesian approach, the parameters are random, and the data are fixed.
- Three ingredients: Prior, likelihood, posterior